Diagnosis and Management of Childhood Tuberculosis

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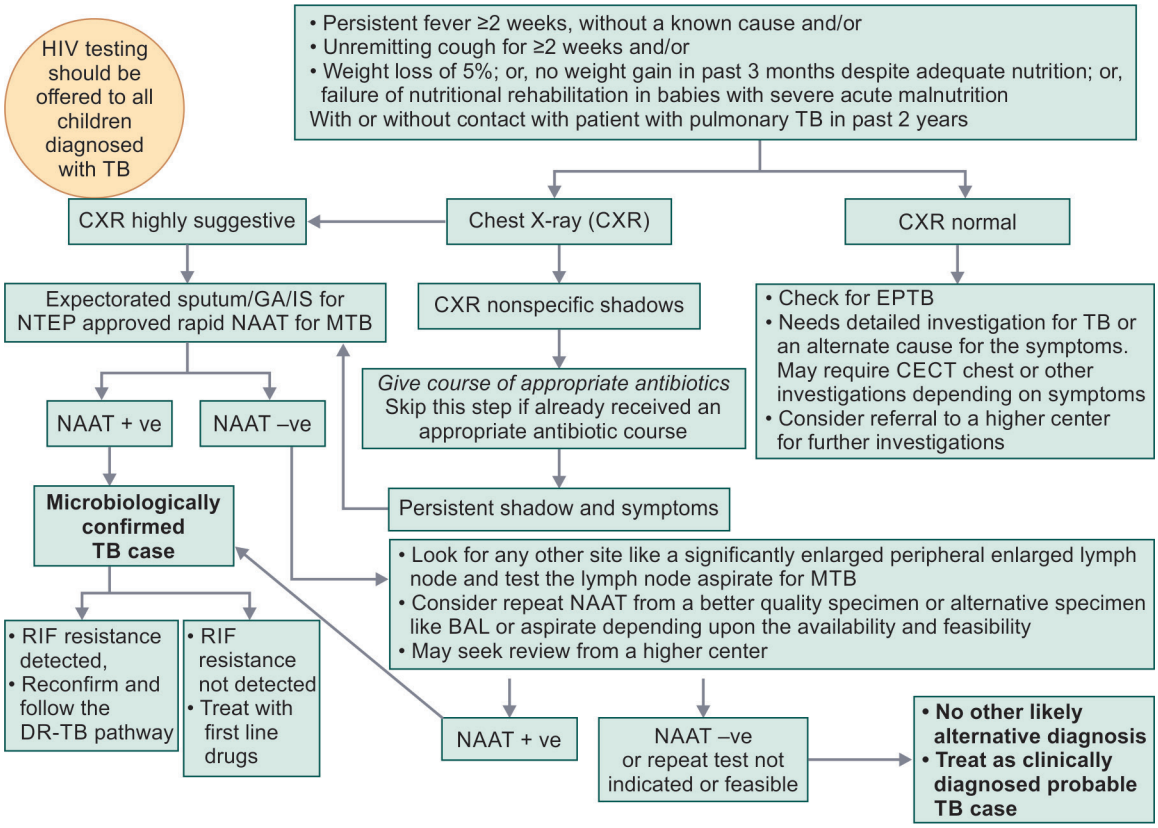
Introduction

- ☑ Tuberculosis (TB) among children is common yet poorly identified disease as it gets both under and overdiagnosed.
- ☑ The newer rapid molecular tests have changed the way TB is diagnosed now, as detailed below.
- ☑ Children with following symptoms, with or without history of contact with an active TB case within last 2 years, are suspected to have TB and are referred to as children with presumptive TB.
- ☑ Persistent and documented fever for 2 weeks or more without a known cause
  - Persistent cough for >2 weeks
  - Significant weight loss of >5% body weight as compared to highest weight recorded in last 3 months
  - Nontender peripheral lymphadenopathy
  - Chronic meningitis with or without neurological deficits

Diagnosis

- ☑ As per the current national strategy, all presumptive TB cases are subjected to National Tuberculosis Elimination Program (NTEP) approved rapid molecular tests on appropriate body specimen for the presence of *Mycobacterium tuberculosis* (MTB) and presence of rifampicin resistance, which is used as a surrogate marker for multidrug-resistant tuberculosis (MDR-TB).
- ☑ This strategy called universal DST or U-DST has been possible by availability of nested nucleic acid amplification test (NAAT) namely Xpert-Rif™ or Truenat® MTB-RIF Dx for both pulmonary and extrapulmonary specimens and line probe assay (LPA or Hain's test) for smear positive clinical specimens or MTB isolates from cultures.
- ☑ To optimize the upfront use of NAAT, the current guidelines suggest to use chest radiograph as the initial screening tool (**Flowchart 1** detailing algorithm for evaluation of a child with pulmonary TB).
- ☑ The main impediment in confirming diagnosis of TB in children is the paucibacillary nature of the disease in addition to difficulty in obtaining adequate, good quality specimen.
- ☑ Furthermore, the low positivity rates of confirmation by molecular or cultures still mandate use of clinicoradiological features for diagnosis. Needless to add that every effort must be made to obtain bacteriological confirmation.

Flowchart 1: Algorithm for pediatric intrathoracic TB among children with no risk factors for drug resistance.



1. Chest X-ray shall be done upfront in cases who are suspected to have TB.
   – If a recent good quality chest X-ray is available, it does not need to be repeated.
2. Highly suggestive chest X-ray refers to miliary shadows, or lymphadenopathy (hilar or mediastinal), or chronic fibrocavitary parenchymal lesions.
3. *Nonspecific chest X-ray:* Refer to patterns other than highly suggestive like consolidations, inhomogneous shadows or bronchopneumonia, etc.
4. NTEP approved NAAT shall be preferred over smear examination in all children.
   – Available NTEP approved NAAT include Xpert Rif, TrueNat, and line probe assay
   – If a specimen is positive by any of these methods, the case is labeled as microbiologically confirmed TB.
   – At the initial step, if self-expectorated sputum is available and imaging/NTEP approved NAAT test is not available or delayed, smear may be done (for ease of availability and low cost).
   – Whenever smear is used for diagnosis at least 2 samples should be tested while a single sample is sufficient for more sensitive NTEP approved NAAT.
   – If a specimen is negative by NTEP approved NAAT (or smear), the second aliquot or a fresh good quality specimen should be submitted for a repeat NAAT and liquid culture.
   – In case of Rif resistance is detected on NAAT, in a new case without any risk factors, a reconfirmation is desirable.
5. Antibiotics of choice include amoxicillin or coamoxiclav.
   – Antibiotics like linezolid or any fluoroquinolone should not be used as they have anti-TB action.
   – In case antibiotic trial has already been given in adequate dose and duration, it may not be repeated.
6. *Clinically diagnosed probable TB case:* Is a patient with a high clinical suspicion for TB disease based on suggestive symptoms, radiology and often supportive circumstances (history of exposure to a TB case) or evidence of infection (positive skin test for TB or positive IGRA). But the rapid microbiological tests are negative. Such a case may be treated as clinically diagnosed patient provided common alternative diagnoses have been ruled out.
   – Where facilities exist, send one aliquot of the specimen for liquid culture, if the NAAT is negative for MTB.

(CECT: contrast-enhanced computed tomography; DR-TB: drug-resistant tuberculosis; HIV: human immunodeficiency virus; MTB: *Mycobacterium tuberculosis*; NAAT: nucleic acid amplification test; NTEP: National Tuberculosis Elimination Program; TB: tuberculosis)

Sample Collection

- ☑ A good sputum sample consists of recently discharged material from the bronchial tree with presence of mucoid or mucopurulent material and should be 2–5 mL in volume.
- ☑ There should be minimum amount of oral or nasopharyngeal material.
- ☑ Self-expectorated sputum sample is the standard microbiological sample for older children.
- ☑ In young children who are unable to produce a good quality sputum specimen, either an induced sputum with 3% hypertonic saline or a gastric lavage is obtained.
- ☑ Bronchoscopy and bronchoalveolar lavage is a very useful but more invasive and specialized investigation, used only when diagnosis is reached with other investigations.

Tests for Bacteriological Confirmation

- ☑ Demonstration of Acid-fast bacillus (AFB) on smear microscopy or culture or by molecular tests is the gold standard of TB diagnosis.
- ☑ Rapid nucleic acid amplification tests like Xpert Rif™/Truenat™ have made it possible to detect MTB with much higher sensitivity as compared to smear and faster than the culture (turnaround time is 2 hours).
- ☑ These tests are also nested for establishing rifampicin resistance. Xpert Rif™/ Truenat™ have become point of care tests for TB diagnosis and detection of rifampicin resistance which is a surrogate marker for MDR-B.
- ☑ Liquid cultures for MTB using the BACTEC/*Mycobacterium* growth indicator tube (MGIT 960) using special liquid ETtubes are preferred over the solid media.
- ☑ Newer generation tests such as Xpert Ultra™ and and Xpert XDR are as yet not approved by NTEP.
- ☑ In addition, NTEP also recommends LPA—which are multiplex NAAT—to test for resistance to rifampicin, isoniazid, and other second-line drugs (fluoroquinolones and second-line injectables).
- ☑ Unlike rapid NAAT, LPA duo its relatively lower sensitivity can be used directly only in smear positive specimens or else after isolating MTB on culture.
- ☑ LPA also has a longer test turnaround time (3–4 days) due to batch testing as compared to individually tested rapid NAAT.

Diagnosis

Imaging Studies

- ☑ *Chest X-ray* is an important imaging modality for pulmonary TB diagnosis in children.
- ☑ Lesions highly suggestive for TB are miliary pattern, chronic fibrocavitatory lesions, and hilar/paratracheal lymph nodes.
- ☑ Consolidations, collapse-consolidation, in homogeneous shadows, or bronchopneumonia are nonspecific patterns.
- ☑ Radiology needs to be correlated with clinical findings.
- ☑ Chest X-ray although is an extremely useful investigation in TB diagnostic is fraught with inter- and intraindividual observer bias.
- ☑ Radiological findings can help differentiate between primary, progressive primary, and reactivation TB.
- ☑ While mediastinal lymphadenopathy, right upper lobe consolidation may be seen in primary TB, collapse consolidation, and necrotic lymph nodes with pressure effects are associated with progressive primary TB.
- ☑ In reactivation TB, cavitatory lesions and endobronchial TB are more common.
- ☑ *Ultrasound* is often used to see deep-seated non-palpable neck nodes and very useful for abdominal imaging.
- ☑ USG is also very useful for performing guided lymph node biopsies.
- ☑ *Contrast CT scan* chest is more accurate in detecting cavitation, bronchial wall thickening, collapse, centrilobular nodules, vertebral access, and tree-in-bud appearance as compared to chest X-ray.
- ☑ Since, CT chest involves radiation and has significant cost compared X-ray, the use is limited to situations where the diagnosis cannot be reached with clinical, microbiological, and chest X-ray findings.

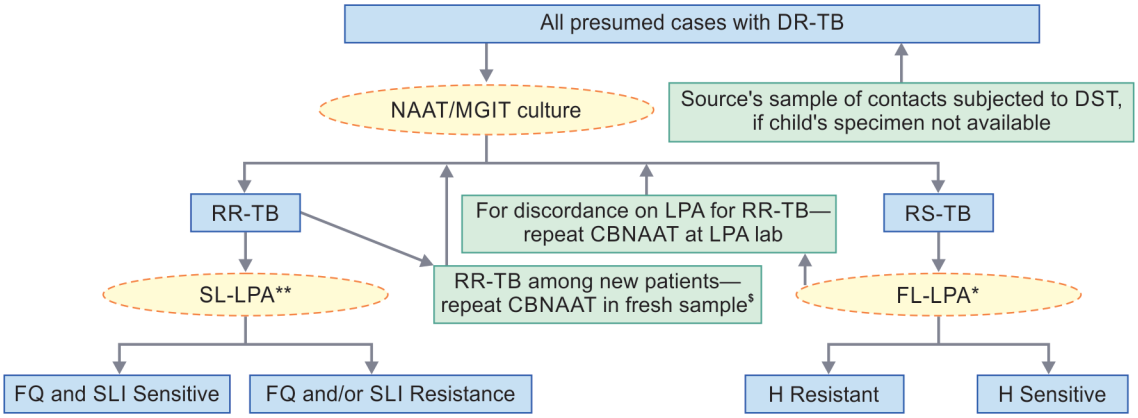
Evaluation of a Child with Suspected Drug Resistance

Children with any of the following should be suspected to have drug resistance:

- ☑ Non or poor response to first-line therapy
- ☑ Treatment after lost to follow-up (defaulter)
- ☑ Recurrent TB (relapse)
- ☑ Contact of MDR patients and/or CLHA and/or with history of TB death in the household:
  - Children suspected to have drug resistance should be carefully evaluated in association with a pediatrician experienced in drug-resistant tuberculosis (DR-TB) or nodal DR-TB center.
  - Microbiological confirmation and drug susceptibility testing is paramount and should always be attempted. The algorithm for evaluation of a child suspected to have DR-TB is summarized in **Flowchart 2**.

Evaluation of a Child with Suspected Drug Resistance

Flowchart 2: Algorithm for evaluation of children with suspected drug-resistant tuberculosis (DR-TB).



*First line LPA (FL-LPA) may be done directly if smear positive; else, send for MGIT followed by FL-LPA to evaluate for R and H resistance.

**Second line LPA (SL-LPA) may be done directly if smear positive; else, send for MGIT followed by SL-LPA or liquid culture DST (Mfx 2.0, Km, Cm, Lzd)

$If rifampicin resistant on repeat test, DRTB regimen is initiated; If repeat test shows Rifampicin resistance not detected or If result is unavailable, DSTB regimen is initiated (FQ: fluoroquinolone; LPA: line probe assay; NAAT: nucleic acid amplification test; MGIT: *Mycobacterium* growth indicator tube; RR-TB: rifampicin-resistant tuberculosis; RS-TB: rifampicin-sensitive tuberculosis)

Treatment of Tuberculosis

- ☑ Standardized therapy of TB with presumed drug sensitivity, followed until recently, has given way to using standardized first-line regimen after ruling out resistance to rifampicin by use of molecular tests discussed above as treating rifampicin-resistant strain with a standardized first-line regimen had a risk of amplification of resistance to more drugs.
- ☑ Therefore, now the therapy is divided as treatment for rifampicin-sensitive TB (RS-TB) and for rifampicin-resistant TB (RR-TB). In cases where the molecular diagnosis fails, the TB cases without any risk factors for drug resistance are initially treated as RS-TB.

Treatment for Rifampicin-Sensitive TB (Tables 1 and 2)

- ☑ TB treatment is biphasic, with an initial intensive phase (IP) aimed at early and rapid killing of MTB thereby preventing death and reducing the infectivity.
- ☑ This is followed by the continuation phase (CP) which eliminates the residual bacilli, thereby reducing the relapses and failures.
- ☑ As there are different types of metabolic populations of MTB in any given patient (extracellular and intracellular, slow and rapidly growing), treatment is done with at least four drugs to target all these populations.
- ☑ Ethambutol is added as the fourth drug in IP and as a third drug in CP due to the high background resistance to isoniazid (INH), up to 12–14%, in our country.

All drugs are preferably administered daily, at once, to achieve peak concentrations in the body.

TABLE 1: Treatment regimen for rifampicin-sensitive TB.

Type of patient[a]	***Regimens***
New microbiologically confirmed RS pulmonary TB	2HRZE + 2HRE[b]
New clinically diagnosed pulmonary TB (probable RS-TB)	
New microbiologically confirmed RS extrapulmonary TB	
New clinically diagnosed extrapulmonary TB (probable RS-TB)	
Drug-sensitive previously treated TB[c] (recurrent, treatment after loss to follow-up, treatment after failure)	

[a] Molecular testing shall be done in all new cases (defined as never had treatment for TB or has taken anti-TB drugs for <1 month) among children with suspected TB at diagnosis and RS-TB cases included in this regimen.

[b] In the case of neuro and spinal TB, the CP is extended to 10 months.

[c] All these categories of children shall be evaluated as DR-TB suspects and assessed as per the DR-TB algorithm. If they are found to be rifampicin and isoniazid sensitive, they shall be restarted on the regimen as for new cases. This group was earlier treated with a category II regimen, which is now withdrawn.

(CP: continuation phase; DR-TB: drug-resistant tuberculosis; RS: rifampicin-sensitive; TB: tuberculosis)

TABLE 2: Drug dosages for rifampicin-sensitive tuberculosis (TB) treatment.

		Range (mg/kg/day)	*Average (mg/kg/day)*	*Maximum dose (mg) for up to 60 kg body weight*
Rifampicin	R	10–20	15	600
Isoniazid	H	7–15	10	300
Pyrazinamide	Z	30–40	35	2,000
Ethambutol	E	15–25	20	1,500
Streptomycin	S	15–20	20	1,000

- ☑ Fixed drug combinations (FDCs) incorporating multidrug therapy for TB with proven bioavailability are available under NTEP and preferred due to their safety, simplified treatment and avoiding errors in missing one or more of the drugs, thus reducing the risk of the emergence of drug-resistant strains.
- ☑ The pediatric FDCs (P) are dispersible and flavored and are three-drug FDC dispersible tablets for IP (H50 + R75 + Z150) and two-drug FDC dispersible tablets (H50 + R75) for CP.
- ☑ Ethambutol is added as a companion drug to both the phases and is available as a 100 mg nondispersible tablet.
- ☑ In addition, for older children nondispersible adult type FDCs (A), available under NTEP, are used (four-drug FDC: H75 + R150 + Z400 + E275; three-drug FDC (H75 + R150 + E275) as detailed in **Table 3**.

TABLE 3: Recommended drug doses and fixed drug combination (FDC) pill combination from 0 to 18 years of age.

Weight band (kg)	*Dose*
4–7	1 P + 1 E
8–11	2 P + 2 E
12–15	3 P + 3 E
16–24	4 P + 4 E
25–29	3 P + 3 E + 1 A
30–39	2 P + 2 E + 2 A
For older children weighing >39 kg, the formulations are used as for adults.	

Adjunctive Therapy along with Anti-TB Drugs

- ☑ *Pyridoxine therapy:* Supplementation with 10 mg/day is recommended for all patients receiving INH-containing regimens.
- ☑ *Steroids in TB:* Steroids decrease inflammation-related injury and shown to reduce morbidity (sequelae) and even mortality in cases of TB when given for appropriate indications.
- ☑ *Definite indications for use of steroids in TB are:*
  - TB meningitis
  - Pericarditis

- Addison's disease
- Miliary TB with alveolocapillary block
- TB uveitis

☑ *Additional indications where steroids may be used:*

- Endobronchial TB
- Bronchial compression
- Mediastinal compression syndrome
- Pleurisy with severe respiratory distress
- Laryngeal TB
- TB immune reconstitution inflammatory syndrome (IRIS)

☑ Prednisolone 1–2 mg/kg/day or dexamethasone 0.6 mg/kg/day or its equivalent are used for 2–4 weeks and then tapered over the next 4 weeks.

Monitoring and Follow-up of Rifampicin-sensitive Pediatric TB Cases:

☑ Clinical follow-up should be done every month during treatment and once every 6 months for 2 years after treatment completion.

☑ An initial visit within 2 weeks of starting the regimen is desirable to reassess that the patient is taking the correct dose, in the correct combination and tolerating all drugs.

☑ *On each clinical follow-up child should be assessed for the following:*

- Improvement in the clinical symptoms
- Physical examination
- Weight monitoring
- Side effects of medications
- Comorbid conditions which may need treatment, e.g., human immunodeficiency virus (HIV), anemia, malnutrition, etc.
- Adherence to treatment should be checked at each follow-up visit.

☑ *Monitoring by investigations is done as follows:*

- *Microbiological:* Respiratory specimens if available are tested at the end of IP and completion of treatment with smear and culture for bacterial negativity.
- *Mycobacterial growth indicator tube (MGIT) culture:* To be performed if the child is not responding even after 4 weeks of therapy. However as most young children are unable to produce sputum or may have complete resolution of symptoms, then the response to treatment may need to be assessed clinically with the help of radiological testing.
- *Follow-up chest radiographs:* To be done only at the end of therapy; or, earlier if clinically indicated due to nonimprovement, deterioration, or emerging complications.

- *Other imaging* such as ultrasonography, echocardiography, and CT/MRI of the affected organ system is also advised at the end of therapy or when the patient is clinically not improving, deterioration or emerging complications.
- *Liver function tests (LFTs):* No routine or baseline LFT test is required for patients on first-line drugs without evidence of hepatopathy. LFTs are done if any child shows symptoms or signs of liver dysfunction.

☑ *Clinical or radiological nonresponse or deterioration during follow-up could be due to:*

- Incorrect diagnosis (particularly in a clinically diagnosed case of TB)
- Lack of adherence to therapy
- Incorrect dosages or drugs taken
- *Inability to retain drugs:* Due to vomiting
- Secondary infection or untreated comorbidity
- Drug resistance
- Paradoxical upgrading reactions (TB-IRIS)

Treatment of Rifampicin-resistant TB in Children:

☑ RR-TB treatment regimens are longer in duration and have higher pill burden and toxicity profile as compared to regimen for RS-TB.

☑ The regimen vary as per severity of disease (shorter course for mild localized pulmonary or nodal disease), and, age [children below 5 and/or weighing <15 kg cannot be given the newer drugs like bedaquiline (BDQ)].

☑ The recommended standardized regimens, for patients meeting the eligibility criteria, are detailed here.

☑ RR cases suitable for short regimen are those who have limited disease, no proven resistance or risk of resistance due to prior prolonged exposure to fluoroquinolones (FQs) and do not have combined InhA and KatG mutations.

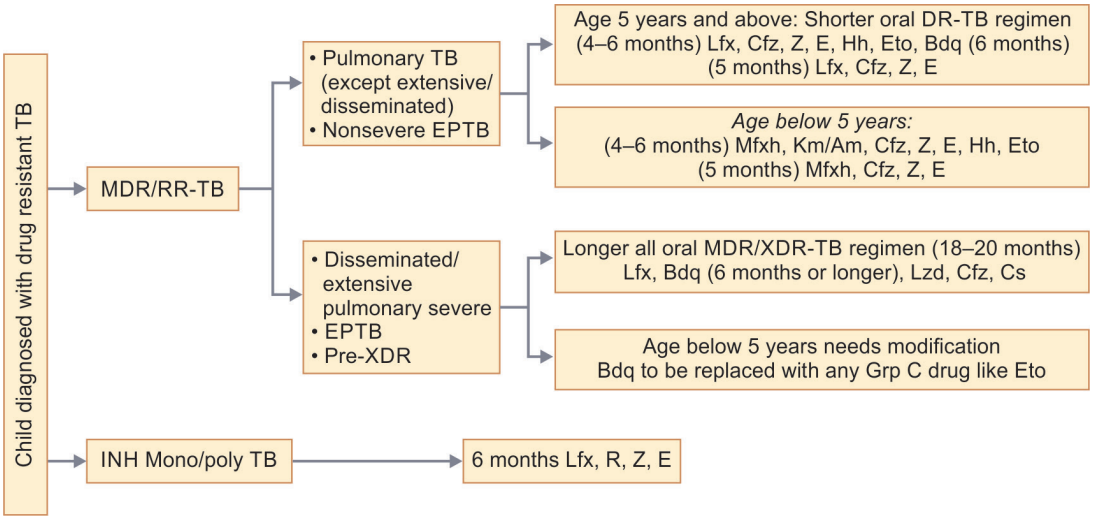
☑ All other patients are prescribed the longer regimen. The standardized longer regimen described below may need modification in cases with additional resistance or prolonged exposure or intolerability to any of the component drug(s). Such cases should be discussed with an expert to device the replacement sequence.

Treatment of INH Monopoly Resistance

☑ Some of the failure and retreatment cases may demonstrate bugs which are rifampicin sensitive but have INH resistance (alone or with any other first-line drug except rifampicin).

☑ Such cases can be treated with 6 months uniphasic levofloxacin containing regimen (6Lfx-RZE), provided they do not have any FQ resistance.

Treatment of Tuberculosis



(Ak: amikacin; Bdq: bedaquiline; Cfz: clofazimine; Cs: cycloserine; Dlm: delanamid; Eto: ethionamide; Hh: high dose INH; Km: kanamycin; Lfx: levofloxacin; MDR: multidrug-resistant; Mfx: moxifloxacin; RR-TB: rifampicin-resistant TB)

Further Reading

☑ National Tuberculosis Elimination Programme (2022). Paediatric TB Management Guideline 2022. [online]. Available from https://tbcindia.gov.in/showfile.php?lid=3668 [Last accessed December, 2022].

Indian Academy of Pediatrics (IAP)

STANDARD TREATMENT GUIDELINES 2022

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